45 Degree Curb Impact Test for Wheels and Tires

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. This test assesses the performance of road wheels and tires (not compact temporary use/spares) in association with a vehicle suspension, structure, and mass using a 100 mm high curb fixture at a 45 degree angle to the direction of vehicle travel.

1.2 Foreword. Two tire and wheel assemblies are assessed in this test per run, and must be installed on the same side of the vehicle.

1.3 Applicability. This test applies to cars, wagons, sport utility vehicles, passenger vans, pickup trucks, and cargo vans. It can be used as a component test or a vehicle system test. Wheels and tires used in Australia, New Zealand, South America, India, and China Market Road Systems.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

None

2.2 GM Standards/Specifications.

GMW15445 GMW16590

2.3 Additional References.

GM Drawing Number 9590710, Drawing Name: Wheel Spec - Wheel Uniformity and Low Point Marking

3 Resources

3.1 Facilities. This test requires a straight, smooth, paved, level test roadway of sufficient length to allow vehicles to accelerate to an impact velocity of at least 64 kph and slow to a stop afterwards. The test roadway should be wide enough to allow the driver to keep the test vehicle on it, if the impact changes the direction of vehicle travel. An anchoring system is necessary to secure the curb fixture to the test roadway.

3.2 Equipment.

3.2.1 Wheel uniformity machinery capable of measuring wheel uniformity coefficients as described in Appendix A, Section A1.

3.2.2 Position measurement machinery capable of determining wheel bead seats radial and lateral coordinates as described in Appendix A, Section A2.

3.2.3 Tire pressure gauge.

3.2.4 Ballast which conforms to appropriate proving ground ballasting procedures.

3.2.5 Radar unit, Velocity Box (Racelogic), or other device capable of correctly indicating vehicle speed.

3.2.6 Curb fixture as specified in Appendix B. The cross-section dimensions are shown in Figure B1.

3.3 Test Vehicle/Test Piece. Test vehicle for which the wheels and tires are intended, or a test vehicle that is as close as possible to design intent. (The test vehicle must have all airbags disabled and verified as disabled by properly trained personnel prior to this severe impact test.)

Test wheels. (If the test is focused on wheel performance, the wheels should be minimum stock condition unless otherwise specified by the test requestor. If the test is focused on tire performance, the wheels may be either minimum stock or nominal stock, and if there are multiple wheel sizes, tests should be performed for all sizes.)

Test tires. (For testing focused on wheel performance, the tires should be new and unused and as close to production intent construction as possible. If there are multiple tire sizes, tests should be performed for all sizes.)

Appropriately sized non-test assemblies for use on the other side of the vehicle from the test assemblies. These wheels and tires should be of the same nominal sizes as the test wheels and tires to ensure that there is no significant vehicle static roll angle due to differences in sizes.

3.4 Test Time.

Calendar time:	2 days
Test hours:	14 hours
Coordination hours:	7 hours

3.5 Test Required Information.

- Tire size(s).
- Tire pressure(s).
- Test assembly locations on the test vehicle, for example, left front and left rear corner locations.
- Test speed, if the test is focused on wheel performance.
- Maximum test speed, if the test is focused on tire performance.

Direction on if or when to perform post-impact vehicle alignment checks and adjustments, if the test is focused on tire performance.

3.6 Personnel/Skills. Personnel should have the following training or credentials:

- Vehicle lift/hoist operation.
- Trained in use of wheel uniformity measurement equipment.
- Trained in use of mounting tire.
- Trained in vehicle alignment.

Must meet appropriate Proving Grounds Driving Regulations to operate a vehicle to and from and on the test roadway.

4 Procedure

4.1 Preparation.

4.1.1 Subparagraph Title. If the test vehicle was not provided with all airbags disabled, they must all be disabled and verified as disabled by properly trained personnel prior to this severe impact test.

4.1.2 Perform Pre-test Assessment of Test Wheels: (The test requestor should be informed of the results in order to determine whether or not a wheel should be accepted or rejected for use. If a wheel does not meet uniformity specifications as noted on its design drawing, its acceptability is to be determined by the test requestor.)

4.1.2.1 Inspect wheels for cracks using dye penetrant or equivalent. If a wheel has a crack, it should not be used as a test specimen.

4.1.2.2 Measure and record pre-test uniformity coefficients as described in Appendix A, Section A1.

4.1.2.3 Measure and record pre-test bead seats radial and lateral coordinates as described in Appendix A, Section A2.

4.1.3 Perform Pre-test Assessment of Test Tires:

4.1.3.1 Inspect for visible damage. If a tire has visible damage, it should not be used as a test specimen.

4.1.4 Mount Each Test Tire on Its Respective Test Wheel.

4.1.5 Perform Pre-test Vehicle Alignment Check and Attempt To Correct To Specification as Necessary. This should be done at curb mass as specified in GMW15445.

4.1.6 Install each test assembly on the test vehicle in the position requested and inflate every tire on the test vehicle to the pressure(s) requested. If position is not specified, typical placement is left front and left rear corner locations. Record the positions of the specific test assemblies. Note that for some vehicles, the front and rear design intent assemblies may differ in their dimensions and pressures, and this intent should be reflected in set up. If the pressures are not specified, use pressure specifications (placard pressure) for the vehicle for which the test assemblies are intended. As noted in 3.3, the non-test wheels and tires should be of the same nominal sizes as the test wheels and tires to ensure that there is no significant vehicle static roll angle due to differences in sizes.

4.1.7 Ballast the test vehicle per appropriate proving ground ballasting procedures and driving regulations: (Start this part of the procedure with the vehicle at curb mass.)

4.1.7.1 Reference: An example of a ballasting procedure is GMW15445 Vehicle Ballast.

4.1.7.2 Weigh the test vehicle prior to ballasting.

4.1.7.3 Calculation of test axle weights: The front test axle weight should be equal to the maximum front axle design load for all vehicle models for which the test assemblies are intended. The rear test axle weight should be equal to 85% of the maximum rear axle design load for all vehicle models for which the test assemblies are intended. Note that the axle loading specified may exceed the design intent placard values for the actual test vehicle used. If the difference is significant, the vehicle manager should be informed.

4.1.7.4 Add ballast to the vehicle to match the calculated test axle weights. It is recommended to have the left side of the vehicle weigh close to the same as the right side of the vehicle. It is difficult in practice to achieve equal weight side to side, so this is not a requirement.

4.1.7.5 Weigh the test vehicle after ballasting. This should include values for total, axles, and corners.

4.1.8 If necessary, calibrate the device selected to indicate vehicle speed to read correctly at the requested impact velocity. For example, if a radar unit is used for this purpose, calibrate it according to its manufacturer's instructions. Note that whenever instrumentation is placed in a test vehicle, its installation and use must be governed by the appropriate proving ground driving regulations.

4.1.9 Configure the curb fixture depending on whether the test is focused on wheel or tire:

4.1.9.1 If the test is to determine wheel performance, the curb fixture must be set up at the appropriate 45 degree angle to the direction of vehicle travel so that the inboard part of the tire and wheel assembly contacts the curb fixture first. This results in the most damage to the wheel, since the inboard rim of the wheel deflects the most under impact. For example, if it is intended to impact the left side assemblies, the curb fixture must be at the 45 degree angle where the left end of it is farther forward than its right end. See Figure 1.

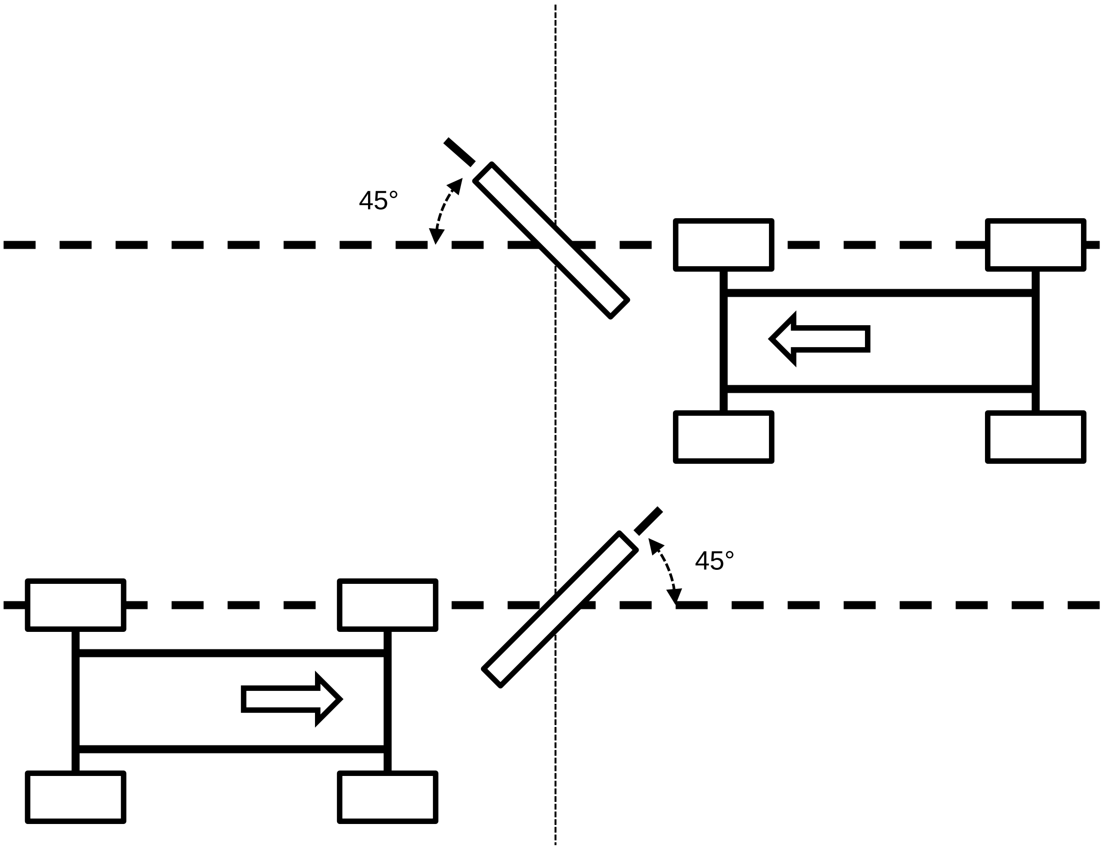

Figure 1: Curb Fixture Orientation for Wheel-Focused Test

4.1.9.2 If the test is to determine tire performance, the curb fixture must be set up at the other 45 degree angle to the direction of vehicle travel (as compared to the wheel test configuration) so that the outboard part of the tire and wheel assembly contacts the curb fixture first. This results in the most damage to the tire, since the outboard rim of the wheel deflects the least under impact, resulting in the greatest impact loading on the tire. For example, if it is intended to impact the left side assemblies, the curb fixture must be at the 45 degree angle where the right end of it is farther forward than its left end. See Figure 2.

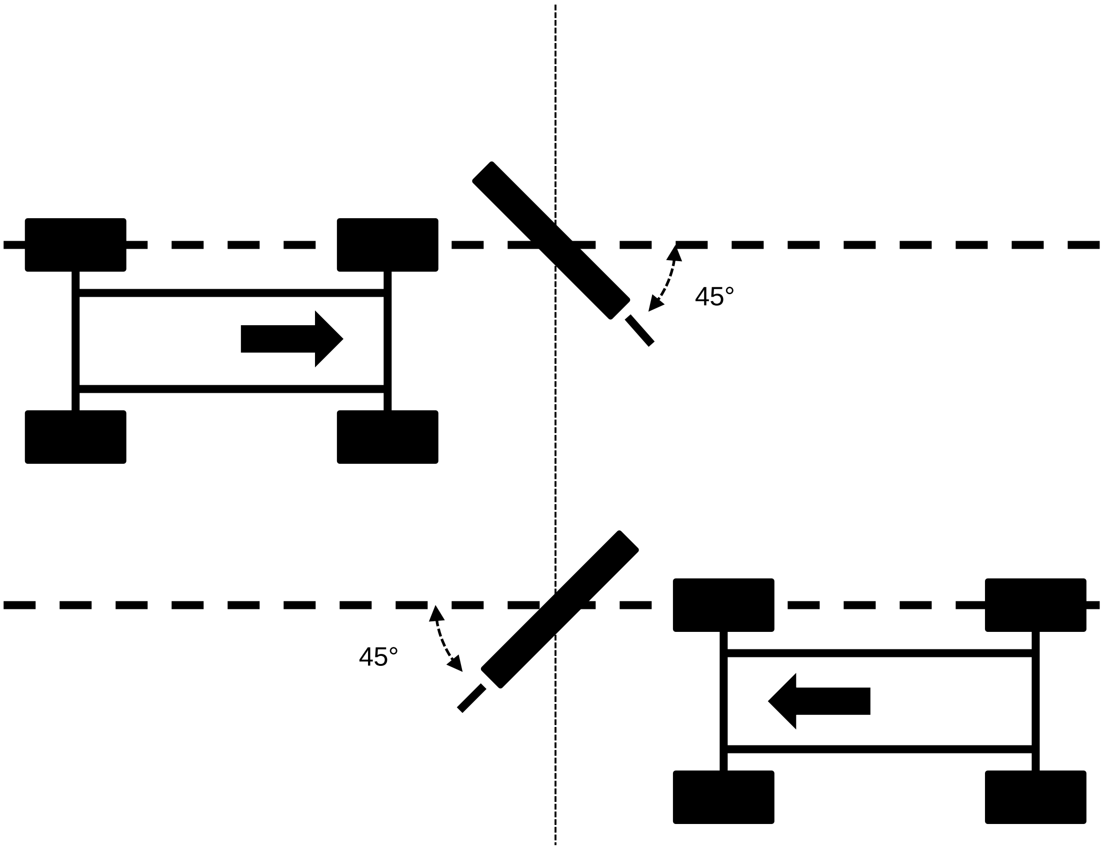

Figure 2: Curb Fixture Orientation for Tire-Focused Test

4.2 Conditions.

4.2.1 Environmental Conditions. The test roadway must be free of snow and ice and any significant loose material, such as gravel or sand. The pavement of the test roadway may be wet, but should not have any standing water.

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 Wheel Test. Perform the following steps, if the test is focused on wheel performance:

4.3.1.1 Drive to the Test Roadway. Coat the impact zone of the curb fixture with a solid paint marker or other suitable material in order to indicate the impact location on the tire. Immediately before driving over the curb fixture, check tire inflation pressures at the closest location allowed, adjust to requested pressure(s), and record. This is done, because in driving to the test facility, tire pressure may change.

4.3.1.2 Drive over the curb fixture at 30 kph or the impact target velocity specified by the test requestor. (Take care to ensure that the velocity of the test vehicle is maintained for the rear impact as well as the front impact, since the front impact may reduce the test vehicle velocity.)

4.3.1.3 Immediately after driving over the curb fixture, check tire pressures at the closest location allowed and record. Record the actual vehicle velocity. Check tire pressures 30 minutes after impacts and record. However, if the ambient temperature is expected to change significantly during the 30 minutes (for example, if tire pressures immediately after the impact are measured in cold weather and then the vehicle is brought back to a heated garage, or, if the weather is hot outside, and the garage is cool) a different method should be used. Allow the vehicle temperature to stabilize, measure initial tire pressures, record, wait 30 minutes, measure the tire pressures again, and record.

4.3.1.4 Remove each test assembly from the test vehicle for post-test evaluation. Inspect each test assembly for damage and record observations. Examine wheels for dents or cracks. Examine tires for any damage, such as bulges or broken plies on the sidewalls or cracks on the tread crown or tire shoulder areas. Photograph as appropriate. Mark the position of the each tire on the wheel before dismounting. Once dismounted, check each wheel for visible deformation on inboard and outboard flanges and record observations. Photograph as appropriate.

4.3.1.5 Perform post-test assessment of test wheels:

4.3.1.5.1 Inspect each wheel for cracks using dye penetrant or equivalent.

4.3.1.5.2 Measure and record post-test uniformity coefficients as described in Appendix A, Section A1.

4.3.1.5.3 Measure and record post-test bead seats radial and lateral coordinates as described in Appendix A, Section A2.

4.3.1.6 Perform post-test assessment of dismounted test tires:

4.3.1.6.1 Inspect for visible damage and record observations.

4.3.1.7 Perform post-test vehicle alignment check and attempt to correct to specification as necessary. This should be done at curb mass as specified in GMW15445.

4.3.2 Tire Test. Perform the following steps, if the test is focused on tire performance:

4.3.2.1 Drive to the Test Roadway. Coat the impact zone of the curb fixture with a solid paint marker or other suitable material in order to indicate the impact location on the tire. Immediately before driving over the curb fixture, check tire inflation pressures at the closest location allowed, adjust to requested pressure(s), and record. This is done, because in driving to the test facility, tire pressure may change.

4.3.2.2 Drive over the curb fixture at 35 kph or the impact target velocity specified by the test requestor. (Take care to ensure that the velocity of the test vehicle is maintained for the rear impact as well as the front impact, since the front impact may reduce the test vehicle velocity.)

4.3.2.3 Immediately after driving over the curb fixture, check tire pressures at the closest location allowed, record, and restore to the requested pressure(s) if necessary. Record the actual vehicle velocity. Examine for tire damage, such as bulges or broken plies on the sidewalls or cracks on the tread crown or tire shoulder areas, and record. Photograph as appropriate. As long as there is no tire damage, repeat the impact described in 4.3.2.2 and the inspection described here in 4.3.2.3, but only for a maximum of four (4) total impacts. Check tire pressures 30 minutes after all impacts are complete and record. However, if the ambient temperature is expected to change significantly during the 30 minutes (for example, if tire pressures immediately after the impact are measured in cold weather and then the vehicle is brought back to a heated garage, or, if the weather is hot outside, and the garage is cool) a different method should be used. Allow the vehicle temperature to stabilize, measure initial tire pressures, record, wait 30 minutes, measure the tire pressures again, and record.

4.3.2.4 Remove each test assembly from the test vehicle for post-test evaluation. Inspect each test assembly for damage and record observations. Examine wheels for dents or cracks. Examine tires for any damage, such as bulges or broken plies on the sidewalls or cracks on the tread crown or tire shoulder areas. Photograph as appropriate. Mark the position of the each tire on the wheel before dismounting. Once dismounted, check each wheel for visible deformation on inboard and outboard flanges and record observations. Photograph as appropriate.

4.3.2.5 Perform post-test assessment of test wheels:

4.3.2.5.1 Inspect each wheel for cracks using dye penetrant or equivalent.

4.3.2.5.2 Measure and record post-test uniformity coefficients as described in Appendix A, Section A1.

4.3.2.5.3 Measure and record post-test bead seats radial and lateral coordinates as described in Appendix A, Section A2.

4.3.2.6 Perform post-test assessment of dismounted test tires:

4.3.2.6.1 Inspect for visible damage and record observations.

4.3.2.7 Perform post-test vehicle alignment check and attempt to correct to specification as necessary. This should be done at curb mass as specified in GMW15445.

5 Data

5.1 Calculations.

5.1.1 Calculate the differences between the pre-test and post-test uniformity coefficients as pre-test values subtracted from the post-test values. These differences are referred to as the changes in uniformity coefficients.

5.1.2 Calculate the differences between the pre-test and post-test bead seat radial and lateral coordinates as post-test values subtracted from the pre-test values. The maximum differences are referred to as the bead seat dent depths.

5.2 Interpretation of Results. See sourcing documents.

5.3 Test Documentation. The test results documentation should include the following:

- Test results report, shown in Appendix D, Data Sheet D1
- Completed Data Sheet, shown in Appendix C
- Wheel uniformity test results report, shown in Appendix D, Data Sheet D2
- Test result data files (Extensible Markup Language (XML) format)
- Wheel uniformity test processing spreadsheet file
- Photos as appropriate

A sample datasheet is given in Appendix C and a sample report is given in Appendix D.

Note: For data being loaded into the Global Tire-Wheel Test System (GTWTS), submit electronically the test results data files (XML format) and other supporting files listed above. GMW16590 defines the electronic file details (e.g., file types, file naming convention, and detailed file formats) for the GTWTS.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.1 Physical Safety. The test vehicle must have all airbags disabled and verified as disabled by properly trained personnel prior to this severe impact test.

6.1.1 Review all appropriate proving grounds driving regulations relevant to this test prior to entering any test road system. Ensure that the test vehicle stays on the test roadway. Due to the nature of this test, it may be necessary to close the test roadway to all other vehicle traffic during testing.

6.1.2 Ergonomic Safety. Do not attempt to lift or move items that are so heavy or resistant that this would cause injury.

6.1.3 Hazardous materials. When hazardous materials are present, take necessary precautions to prevent exposure above safe limits.

7 Notes

7.1 Glossary. Not applicable.

7.2 Acronyms, Abbreviations, and Symbols.

GTWTS	Global Tire-Wheel Test System
ID	Identification
kph	kilometers per hour
P/N	Part Number
PQMS	Program Quality Management System
TIR	Total Indicated Runout
TR	Test Request
XML	Extensible Markup Language

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW16768

9 Release and Revisions

This standard was originated in January 2012. It was first approved by the Tire and Wheel Global Subsystem Leadership Team in November 2013. It was first published in November 2013.

Issue	Publication Date	Description (Organization)
1	November 2013	Initial publication.

Appendix A

A1 Wheel Uniformity Coefficients

The equipment used to assess wheel uniformity should be able to make at least 256 measurements per wheel revolution. The wheel uniformity coefficients are as follows:

A1.1 Average radial first harmonic runout of the wheel bead seats about the bolt circle and the center pilot. **Note**: For an example of these measurements, refer to Drawing 9590710, Specification 5.1.9, Section 2.1.

A1.2 Average radial second harmonic runout of the wheel bead seats about the bolt circle or the center pilot. **Note**: The second harmonic runout is the same to within the fast fourier transform mathematical precision regardless of center of rotation. See paragraph A1.1 for reference.

A1.3 Average radial total indicated runout (TIR) of the bead seats about the bolt circle and the center pilot.

A1.4 Radial TIR of the inboard bead seat about the bolt circle and the center pilot.

A1.5 Radial TIR of the outboard bead seat about the bolt circle and the center pilot.

A1.6 Radial first harmonic runout between the bolt circle and the center pilot. **Note:** For an example of these measurements, refer to Drawing 9590710, Specification 5.1.9, Section 3.3.

A1.7 Average Lateral TIR of the wheel bead seats about the bolt circle or the center pilot. **Note:** For an example of these measurements, refer to Drawing 9590710, Specification 5.1.9, Section 2.2.

A2 Wheel Bead Seats Radial and Lateral Coordinates

Wheel bead seats radial and lateral coordinates are the raw data that are used to calculate the wheel uniformity measurements discussed above. A wheel uniformity machine will typically run probes around the wheel, in contact with its bead seats, in order to obtain these coordinates. If the uniformity machine is configured to record these coordinates it may be used to make the following measurements:

A2.1 Radial coordinates of the bead seats.

A2.2 Lateral coordinates of the bead seats.

Appendix B

B1 Curb Fixture. The curb fixture is to be at least 1000 mm in length, 100 mm in height, and at least 100 mm in width. It is to have a particular cross-section shape, as shown in Figure B1.

B1.1 Length: At least 1000 mm long. The length is not critical, but typically should not exceed the track width of the test vehicle, and ideally would be less than the length between the inboard surfaces of the test vehicle front or rear tires, whichever are closer together.

B1.2 Height: 100 mm in height, as shown in Figure B1.

B1.3 Width: The width must be at least 100 mm, but should not exceed 150 mm.

B1.4 Cross-Section Shape: The impact side of the curb fixture must have the dimensions as shown in Figure B1, which specifies the chamfer and blend details. Note that Figure B1 shows a symmetric design, where either side may be used for impacts. The corners of the chamfers (indicated with arrows from the "Blends with radii 15 mm" notation) are to have blends, although the drawing does not show those blends. Note also that the chamfer dimensions are 25 mm high by 30 mm wide to be consistent with the 75 mm ground-to-chamfer height dimension shown.

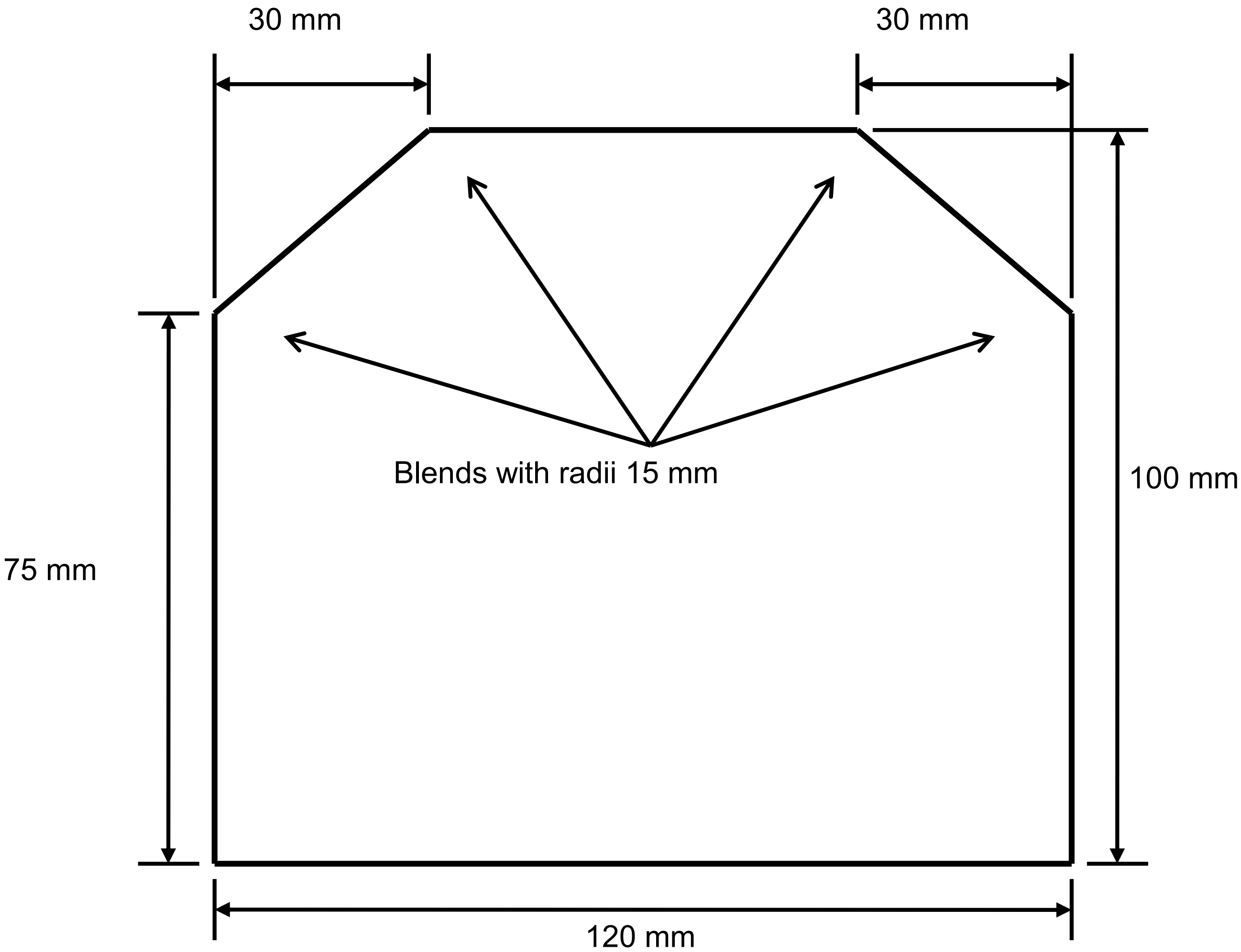

Figure B1: Cross-Section Dimensions

Appendix C

Sample Test Data Sheet (Page 1 of 3)

45 Degree Curb Impact Test for Wheels and Tires Datasheet

Tire-Wheel Systems

Test Request Number:

PQMS Activity Number:

TEST REQUEST DETAILS		
Model Year / Program:		**Requester:**
Test Purpose:		**Requester's Phone:**
WHEEL PART DETAILS		
Supplier:	**Attachment:**	**Part Number (P/N):**
Wheel Diameter / Rim Size:	**Wheel Nut P/N:**	**Chart Number:**
Offset:	**Broadcast Code:**	**Wheel Finish:**
Material Rim / Disc:	**Wheel Bolt P/N:**	**Steel Gauge Rim / Disc (mm):**
TEST CONDITIONS		
Test Impact Condition:		**Gross Vehicle Weight (GVW) (kg):**
Number of Impacts:		**Gross Axle Weight Rating Front (kg):**
Wheel Test Speed (km/h):		**Gross Axle Weight Rating Rear (kg):**
Tire Test Initial Speed km/h):		**Unsprung Mass of Front Corner (kg):**
Tire Test Maximum Speed (km/h):		**Unsprung Mass of Rear Corner(kg):**
Wheel Nut Torque (Nm):		**Wheel Rim Construction and Flange Type:**
TR Comment:		
TIRE PART DETAIL		
Tire Position:		**Tire Supplier:**
Tire Size:		**Trade Name:**
Part #:		**Tread Type:**
Construction Number:		**Requested Tire Pressure (kPa):**
Load Index / Speed Rating:		**Flange Protector Description:**
VEHICLE DETAIL		

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GM Confidential

Sample Test Data Sheet (Page 2 of 3)

45 Degree Curb Impact Test for Wheels and Tires Datasheet

Tire-Wheel Systems

Test Request Number:

PQMS Activity Number:

Vehicle Number: ____________________ **Vehicle Model:** ____________________
Vehicle Make: ____________________ **Model Year:** ____________________

TEST RESULTS

LF Corner Load Before Ballasting (kg): ____________________ **LR Corner Load Before Ballasting (kg):** ____________________
RF Corner Load Before Ballasting (kg): ____________________ **RR Corner Load Before Ballasting (kg):** ____________________
Test LF Corner Load (kg): ____________________ **Test LR Corner Load (kg):** ____________________
Tested RF Corner Load (kg): ____________________ **Tested RR Corner Load (kg):** ____________________
Tested Front Pressure (kPa): ____________________ **Tested Rear Pressure (kPa):** ____________________
Tester / Test Site: GMNA-TWS / ______________ **Pre-Test Alignment Performed (Y/N)?:** Y_____ N______
Number of Wheels: 4 **Velocity Measurement Device Calibrated (Y/N)?:** Y_____ N______
Test Date: ____________________ **Description of Curb Fixture Orientation:** ____________________
Technician: ____________________ ____________________

Wheel ID: ________________ **Pre-Test Rim Edge/Flange Coordinates Measured?:** Y_____ N______
Pre-Test Inspection?: Y_____ N______ **Tire Mounted and Marked With TR and ID of This Wheel?:** Y_____ N______
Cracks Found?: Y_____ N______ **Wheel Position (LF, LR, LRI, LRO, RF, RR, RRI, RRO):** ____________________
Photos Taken?: Y_____ N______ **Tire Pre-Test Description:** ____________________
Pre-Test Uniformity?: Y_____ N______ ____________________

List Velocity For Front Impact:	
List Velocity For Rear Impact:	
Vehicle Drivable After Impact:	**If No, Which Impact:**
Total Sudden Loss of Air From Assembly/Assemblies:	**If Yes, Which Assembly/Assemblies:**
Assembly/Assemblies Maintained Air Pressure:	**If No, which Assembly/Assemblies:**
Assembly/Assemblies Air Pressure(s) Immediately After Test (kPa):	

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Sample Test Data Sheet (Page 3 of 3)

45 Degree Curb Impact Test for Wheels and Tires Datasheet

Tire-Wheel Systems

Test Request Number:

PQMS Activity Number:

Assembly/Assemblies Air Pressure(s) 30 Minutes After Test (kPa):
Assembly/Assemblies Impact Location(s) Marked on Tire(s) & Wheel(s):

Wheel ID: ________________

Post-Test Inspection?: Y_____ N______

Cracks Found?: Y_____ N______

Photos Taken?: Y_____ N______

Post-Test Uniformity?: Y_____ N______

Post-Test Rim Edge/Flange Coordinates Measured?: Y_____ N______

Tire Mounted and Marked With TR and ID of This Wheel?: Y_____ N______

Wheel Position (LF, LR, LRI, LRO, RF, RR, RRI, RRO): _____________________

Tire Post-Test Description: _____________________

____

Post-Test Alignment Performed (Y/N)?: Y_____ N______

Additional Comments: ____

____

____

Date Entered Into GTWTS: ___________________________ **Entered By:** ________________________________

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Appendix D

Data Sheet D1: Sample Test Results Reports (Page 1 of 4)

45 Degree Curb Impact Test for Wheels and Tires Report

Tire-Wheel Systems

Test Request Number:
PQMS Activity Number:

TEST REQUEST DETAILS		
Model Year / Program:		**Requester:**
Test Purpose:		**Requester's Phone:**
WHEEL PART DETAIL		
Supplier:	**Attachment:**	**Part Number (P/N):**
Wheel Diameter / Rim Size:	**Wheel Nut P/N:**	**Chart Number:**
Offset:	**Broadcast Code:**	**Wheel Finish:**
Material Rim / Disc:	**Wheel Bolt P/N:**	**Steel Gauge Rim / Disc (mm):**

TEST CONDITIONS	
Test Impact Condition:	**Gross Vehicle Weight (GVW) (kg):**
Number of Impacts:	**Gross Axle Weight Rating Front (kg):**
Wheel Test Speed (km/h):	**Gross Axle Weight Rating Rear (kg):**
Tire Test Initial Speed (km/h):	**Unsprung Mass of Front Corner (kg):**
Tire Test Maximum Speed (km/h):	**Unsprung Mass of Rear Corner(kg):**
Wheel Nut Torque (Nm):	**Wheel Rim Construction and Flange Type:**
TR Comment:	
TIRE PART DETAIL	
Tire Position:	**Tire Supplier:**
Tire Size:	**Trade Name:**
Part #:	**Tread Type:**
Construction Number:	**Requested Tire Pressure (kPa):**
Load Index / Speed Rating:	**Flange Protector Description:**
VEHICLE DETAIL	
Vehicle Number:	**Model:**
Make:	**Year:**

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Data Sheet D1: Sample Test Results Reports (Page 2 of 4)

45 Degree Curb Impact Test for Wheels and Tires Report

Tire-Wheel Systems

Test Request Number:
PQMS Activity Number:

TEST RESULTS

LF Corner Load Before Ballasting (kg):	**LR Corner Load Before Ballasting (kg):**
RF Corner Load Before Ballasting (kg):	**RR Corner Load Before Ballasting (kg):**
Tested LF Corner Load (kg):	**Tested LR Corner Load (kg):**
Tested RF Corner Load (kg):	**Tested RR Corner Load (kg):**
Tested Front Pressure (kPa):	**Tested Rear Pressure (kPa):**
Tester / Site:	**Pre-Test Alignment Performed (Y/N)?:**
TIP Approved:	**Velocity Measurement Device Calibrated (Y/N)?:**
Number of Wheels:	**Description of Curb Fixture Orientation:**
Test Date:	
Technician:	
Wheel ID:	**Pre-Test Rim Edge/Flange Coordinates Measured?:**
Pre-Test Inspection?:	**Tire Mounted and Marked With TR and ID of This Wheel?:**
Cracks Found?:	**Wheel Position:**
Photos Taken?:	**Tire Pre-Test Description:**
Pre-Test Uniformity?:	
Wheel ID:	**Pre-Test Rim Edge/Flange Coordinates Measured?:**
Pre-Test Inspection?:	**Tire Mounted and Marked With TR and ID of This Wheel?:**
Cracks Found?:	**Wheel Position:**
Photos Taken?:	**Tire Pre-Test Description:**
Pre-Test Uniformity?:	

List Velocity For Front Impact:		
List Velocity For Rear Impact:		

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Data Sheet D1: Sample Test Results Reports (Page 3 of 4)

45 Degree Curb Impact Test for Wheels and Tires Report

Tire-Wheel Systems

Test Request Number:

PQMS Activity Number:

Vehicle Drivable After Impact:		**If No, Which Impact:**	
Total Sudden Loss of Air From Assembly/Assemblies:		**If Yes, Which Assembly/Assemblies:**	
Assembly/Assemblies Maintained Air Pressure:		**If No, which Assembly/Assemblies:**	
Assembly/Assemblies Air Pressure(s) Immediately After Test (kPa):			
Assembly/Assemblies Air Pressure(s) 30 Minutes After Test (kPa):			
Assembly/Assemblies Impact Location(s) Marked on Tire(s) & Wheel(s):			

Wheel ID:
Post-Test Inspection?
Cracks Found?
Photos Taken?
Post-Test Uniformity?

Post-Test Rim Edge/Flange Coordinates Measured?:
Crack Location Codes:
Crack Location Comment:
Description of Wheel at End of Test:
Description of Tire at End of Test:

Wheel ID:
Post-Test Inspection?
Cracks Found?
Photos Taken?
Post-Test Uniformity?

Post-Test Rim Edge/Flange Coordinates Measured?:
Crack Location Codes:
Crack Location Comment:
Description of Wheel at End of Test:
Description of Tire at End of Test:

Post-Test Alignment Performed (Y/N)?:

Additional Comments:

GM Proprietary Page 3 of 4 Date Printed:

Data Sheet D1: Sample Test Results Reports (Page 4 of 4)

45 Degree Curb Impact Test for Wheels and Tires Report

Tire-Wheel Systems

Test Request Number:

PQMS Activity Number:

Test Number	Tester	Test Date	Comments	Result Status		
				Approval	Name	Date

FILE ATTACHMENTS

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Data Sheet D2: Sample Wheel Uniformity Test Results Report (Page 1 of 3)

W01WU1 Uniformity Report

Tire-Wheel Systems

Test Request Number: WYY_NNNNN
PQMS Activity Number:

TEST REQUEST DETAILS

Model Year / Program:	**Requester:**
Test Purpose:	**Requester's Phone:**

WHEEL PART DETAIL

Supplier:	**Attachment:**	**Part Number (P/N):**
Wheel Diameter / Rim Size:	**Wheel Nut P/N:**	**Chart Number:**
Offset:	**Wheel Nut Finish:**	**Broadcast Code:**
Wheel Finish:	**Wheel Bolt P/N:**	**Steel Gauge Rim / Disc (mm):**
Material Rim / Disc:	**Wheel Bolt Finish:**	

TEST CONDITIONS

TR Comment:

TEST RESULTS

Wheel ID:		**Wheel Test:**	curb
Tester / Site:		**Wheel Test Other Description:**	
Machine Name / ID:		**TIP Approved:**	N/A
COMMENT:	Bolt Average Radial TIR pre -.--- post -.--- delta -.---; Pilot Avg. Rad. TIR pre -.--- post -.--- delta -.---. (Calculated from probe display data.)		

UNIFORMITY COEFFICIENTS (mm):	Pre-Test:	Post-Test:	Delta:
Bolt Average Radial 1st Harmonic Magnitude:			
Pilot Average Radial 1st Harmonic Magnitude:			
Bolt or Pilot Average Radial 2nd Harmonic Magnitude:			
Bolt To Pilot Hole Magnitude:			
Pilot Average Lateral TIR:			

BEAD SEAT DENT DEPTHS (mm):	Method 1:	Method 2:
Inboard Radial Bead Seat:		
Outboard Radial Bead Seat:		
Inboard Lateral Bead Seat:		

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Data Sheet D2: Sample Wheel Uniformity Test Results Report (Page 2 of 3)

W01WU1 Uniformity Report

Tire-Wheel Systems

Test Request Number: WYY_NNNNN

PQMS Activity Number:

Outboard Lateral Bead Seat:			
RIM EDGE / FLANGE DENT DEPTHS (mm):		**Method 1:**	**Method 2:**
Inboard Radial Rim Edge / Flange:			
Outboard Radial Rim Edge / Flange:			
Inboard Lateral Rim Edge / Flange:			
Outboard Lateral Rim Edge / Flange:			
Pre-Test Technician:		**Pre-Test Date:**	DD MMM YYYY
Post-Test Technician:		**Post-Test Date:**	05 MAY 2011

Wheel ID:		**Wheel Test:**	curb
Tester / Site:		**Wheel Test Other Description:**	
Machine Name / ID:		**TIP Approved:**	
COMMENT:	Bolt Average Radial TIR pre -.— post -.— delta -.—; Pilot Avg. Rad. TIR pre -.— post -.— delta -.—. (Calculated from probe display data.)		

UNIFORMITY COEFFICIENTS (mm):	**Pre-Test:**	**Post-Test:**	**Delta:**
Bolt Average Radial 1st Harmonic Magnitude:			
Pilot Average Radial 1st Harmonic Magnitude:			
Bolt or Pilot Average Radial 2nd Harmonic Magnitude:			
Bolt To Pilot Hole Magnitude:			
Pilot Average Lateral TIR:			

BEAD SEAT DENT DEPTHS (mm):	**Method 1:**	**Method 2:**
Inboard Radial Bead Seat:		
Outboard Radial Bead Seat:		
Inboard Lateral Bead Seat:		
Outboard Lateral Bead Seat:		
RIM EDGE / FLANGE DENT DEPTHS (mm):	**Method 1:**	**Method 2:**

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Data Sheet D2: Sample Wheel Uniformity Test Results Report (Page 3 of 3)

W01WU1 Uniformity Report

Test Request Number: WYY_NNNNN

PQMS Activity Number:

Tire-Wheel Systems

Inboard Radial Rim Edge / Flange:			
Outboard Radial Rim Edge / Flange:			
Inboard Lateral Rim Edge / Flange:			
Outboard Lateral Rim Edge / Flange:			
Pre-Test Technician:		**Pre-Test Date:**	DD MMM YYYY
Post-Test Technician:		**Post-Test Date:**	DD MMM YYYY

Wheel ID	Tester	Test Date	Comment	Result Status		
				Approval	Name	Date

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